

Question

Four fraud detection models are evaluated:

Model	Precision	Recall	F1
A	0.80	0.22	0.34
B	0.55	0.71	0.62
C	0.77	0.41	0.54
D	0.61	0.36	0.45

Which model catches the largest percentage of actual fraud cases?

- A. A
- B. B
- C. C
- D. D

Answer: B

Model B

Explanation

Recall (also called sensitivity or true positive rate) measures:

$$\text{Recall} = \frac{TP}{TP + FN}$$

“Of all the *actual* fraud cases, what fraction did the model catch?”

Comparing the recall column:

- A: 0.22
- B: 0.71 ← highest
- C: 0.41

Question

Which model produces the highest percentage of correct fraud alerts once it flags a transaction?

Model	Precision	Recall
A	0.91	0.18
B	0.62	0.70
C	0.71	0.52
D	0.68	0.44

- A. A
- B. B
- C. C
- D. D

Answer: A

Model A

Explanation

Precision answers:

$$\text{Precision} = \frac{TP}{TP + FP}$$

“Of all the transactions the model *flagged* as fraud, how many were actually fraudulent?”

The phrase “once it flags a transaction” is the key: it conditions on the *prediction* being positive. That is exactly precision.

Comparing the precision column:

- A: 0.91 ← highest
- B: 0.62
- C: 0.71
- D: 0.68

Question

A loan portfolio contains 98% good loans and 2% defaults.
A model predicts every loan as non-default.
What is the model's approximate accuracy?

- A. 2%
- B. 50%
- C. 98%
- D. 100%

Answer: C

98%

Explanation

Accuracy is:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

If the model predicts *non-default* for every loan:

It correctly classifies all 98% good loans (true negatives).

It incorrectly classifies all 2% defaults as good (false negatives).

$$\text{Accuracy} = \frac{98\%}{100\%} = 98\%$$

The trap: 98% accuracy sounds excellent, but the model has *zero* recall for defaults—it catches *none* of the actual defaults. This is the classic class imbalance problem: a trivial majority-class predictor can achieve high accuracy while being completely useless for the minority (and usually more important) class. This is why accuracy alone is a poor metric for imbalanced problems, and why precision, recall, and F1 are preferred.

Question

Which metric combines precision and recall into a single measure?

- A. Accuracy
- B. FPR
- C. F1 Score
- D. AUC

Answer: C

F1 Score

Explanation

The F1 score is the harmonic mean of precision and recall:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Why harmonic mean and not arithmetic mean? The harmonic mean punishes extreme values. If a model has precision = 1.0 and recall = 0.0, the arithmetic mean would be 0.5, but the harmonic mean is 0. So F1 is only high when *both* precision and recall are reasonably high—it forces a balance.

Why the others are wrong:

Accuracy: measures overall correctness, not a precision/recall combination.

FPR: false positive rate, a single error rate.

AUC: area under the ROC curve, a threshold-independent ranking measure.

Question

Which model provides the best balance between precision and recall?

Model	Precision	Recall	F1
A	0.90	0.15	0.26
B	0.60	0.70	0.65
C	0.75	0.30	0.43
D	0.55	0.40	0.46

- A. A
- B. B
- C. C
- D. D

Answer: B

Model B

Explanation

“Best balance between precision and recall” means the highest F1 score, since F1 is the harmonic mean of the two.

Comparing the F1 column:

- A: 0.26 (high precision 0.90 but terrible recall 0.15)
- B: 0.65 ← highest
- C: 0.43
- D: 0.46

Model B has the highest F1 (0.65), meaning it achieves the best trade-off between catching positives (recall 0.70) and being correct when it predicts positive (precision 0.60).

Note: Model A has the highest precision (0.90) but the *worst* F1, because its recall is only 0.15. This illustrates why you should not judge a model by a single metric—high precision alone does not mean a good balance.

Question

An ROC curve plots:

- A. Precision against Recall
- B. TPR against FPR
- C. Accuracy against Threshold
- D. Recall against Accuracy

Answer: B

TPR against FPR

Explanation

An ROC curve (Receiver Operating Characteristic) plots:

True Positive Rate (TPR) vs. False Positive Rate (FPR)

across all possible classification thresholds.

Where:

$$\text{TPR} = \frac{TP}{TP + FN} = \text{Recall} / \text{Sensitivity}$$

$$\text{FPR} = \frac{FP}{FP + TN} = 1 - \text{Specificity}$$

How to read it:

The top-left corner (TPR = 1, FPR = 0) is perfect classification.

The diagonal line (TPR = FPR) represents random guessing.

A curve closer to the top-left corner is a better model.

Question

Two models have AUC values:

$$AUC_A = 0.94$$

$$AUC_B = 0.72$$

Which statement is most accurate?

- A. Model B separates classes better.
- B. Model A separates classes better.
- C. Both models perform identically.
- D. AUC measures precision.

Answer: B

Model A separates classes better.

Explanation

AUC (Area Under the ROC Curve) measures the model's ability to *discriminate* between positive and negative classes.

Interpretation: AUC is the probability that a randomly chosen positive example is ranked higher (scored more positively) than a randomly chosen negative example.

Range and meaning:

AUC = 0.5: random guessing (no discrimination)

AUC = 1.0: perfect separation

AUC = 0.94 (Model A): excellent discrimination

AUC = 0.72 (Model B): acceptable discrimination

Since $AUC_A = 0.94 > AUC_B = 0.72$, Model A separates the classes better.

Why the others are wrong:

A: Model B has the *lower* AUC, so it separates worse.

Question

A bank lowers its approval threshold for identifying risky loans.
What is the most likely outcome?

- A. Lower recall and lower FPR
- B. Higher recall and higher FPR
- C. Higher precision and lower recall
- D. Lower precision and lower recall

Answer: B

Higher recall and higher FPR

Explanation

Lowering the threshold means the model becomes more eager to predict the positive class ("risky loan"). It flags more loans as risky.

Effect on recall (TPR):

More loans are flagged positive.

More of the truly risky loans get caught.

⇒ Recall increases.

Effect on FPR:

More loans are flagged positive overall.

More *safe* loans get incorrectly flagged as risky.

⇒ FPR increases.

This is the fundamental threshold trade-off: lowering the threshold increases both TPR and FPR. You cannot increase recall without also increasing false positives (unless the model is perfect).

Why the others are wrong:

A: Recall does not decrease when you lower the threshold.

C: Precision typically *decreases* (more false positives), and recall *increases*, not the reverse.

D: Recall increases, not decreases.

Question

A model incorrectly flags 200 legitimate transactions out of 5,000 legitimate transactions.
The FPR is closest to:

- A. 1%
- B. 4%
- C. 20%
- D. 40%

Answer: B

4%

Explanation

False Positive Rate (FPR) is:

$$FPR = \frac{FP}{FP + TN}$$

Here:

$FP = 200$ (legitimate transactions incorrectly flagged)

$TN = 5000 - 200 = 4800$ (legitimate transactions correctly not flagged)

Total legitimate transactions (actual negatives) = 5000

$$FPR = \frac{200}{5000} = 0.04 = 4\%$$

Why the others are wrong:

A (1%): Would require only 50 false positives.

C (20%): Would require 1,000 false positives.

D (40%): Would require 2,000 false positives.

Question

Which pairing is correct?

- A. Precision = Of actual positives, how many were found?
- B. Recall = Of predicted positives, how many were correct?
- C. Precision = Of predicted positives, how many were correct?
- D. FPR = Of actual positives, how many were missed?

Answer: C

Precision = Of predicted positives, how many were correct?

Explanation

The correct definitions:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (\text{of predicted positives, how many are correct?})$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (\text{of actual positives, how many are found?})$$

Why the others are wrong:

A: "Of actual positives, how many were found?" is the definition of recall, not precision.

B: "Of predicted positives, how many were correct?" is the definition of precision, not recall. (The pairing is swapped.)

D: "Of actual positives, how many were missed?" is the false negative rate ($FN/(TP + FN) = 1 - \text{recall}$), not FPR. FPR is "of actual *negatives*, how many were incorrectly flagged?"

Memory aid:

Metric	Conditions on
Precision	Predicted positives ($\hat{Y} = 1$)
Recall	Actual positives ($Y = 1$)
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Model A

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Comparing the precision column:

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- D: 0.68